ActDelta: Decision Relevance Is Learnable Offline but Does Not Yet Beat Periodic Sampling Online

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Abstract

Edge-offloaded robot policies pay a recurring camera-uplink cost on every control step, and compression reduces the bytes without answering which observations are worth sending at all. We test *decision relevance*—the action change caused by replacing a true observation with its prediction—as an alternative to reconstruction fidelity, and evaluate it in closed loop rather than offline only.

Across four LIBERO suites and seven horizons, four fidelity metrics correlate weakly with action divergence on 1,400 held-out pairs (pooled Spearman −0.117 to −0.028) and disagree with its median-threshold decision on 48.9–53.9% of frames, while a frozen latent head predicting divergence from robot-side state alone reaches Spearman 0.684 and AUROC 0.833 with no online policy call. This offline gain does not transfer. Against rate-matched periodic sampling in an 800-cell blind test, the learned trigger changes success by +3.13 pp at a 25% budget and −3.75 pp at 50%, with both intervals covering zero; a pre-registered oracle handed the true divergence fails at the same scale, so the fault lies in the target rather than the estimator. Bernoulli sampling is not separated from the learned trigger across the sweep. Threshold triggering instead creates a heavier observation-age tail, and episode outcomes separate on age far more sharply than on instantaneous divergence. Decision relevance is learnable offline but is the wrong quantity to schedule on.

CCS Concepts

• **Networks** → **Network measurement**; • **Computer systems organization** → *Robotics.*

Keywords

Predictive transmission, event-triggered sampling, world models, vision-language-action models, edge offloading, task-oriented communication, measurement, negative results

1 Introduction

Generalist robot policies are large. A contemporary vision-language-action model carries billions of parameters and expects several camera views per control step, which is more than a mobile manipulator can host [4, 21, 38]. The standard deployment therefore splits the loop across the network: the robot uploads an observation, an edge server runs the policy, and a chunk of future actions returns while the previous chunk is still executing [8, 12, 19, 24, 37]. The asymmetry is severe: the downlink carries a few hundred numbers, the uplink carries images on every step for the length of the episode. Compression shrinks each frame but does not answer a prior question—which frames are worth sending at all (Figure 1)?

Networked control has a standard answer. Event-triggered and send-on-delta schemes hold a shared predictor at both ends and transmit only once the receiver's estimate has drifted too far [18, 30, 35, 40]. The rule is fidelity-centered: send when reconstruction error grows. That is the right rule when the receiver is an estimator, because estimate error is what an estimator is judged on. A robot policy is not an estimator. It consumes an observation in order to act, and its action is a sharply non-linear function of the frame: a large visual change can leave the next action chunk untouched, while a small one—a gripper closing on a contact edge—can change it completely. Whether reconstruction error even correlates with the change that matters is an empirical question, and it has not been measured for a modern policy.

We measure it and then act on the answer. *Decision relevance* is the action change caused by substituting a predicted observation for the true one: run the frozen policy twice under common noise, once on each, and take the RMS difference of the emitted action chunks. This is a property of the policy, not of the image. ActDelta learns it offline from robot-side state alone—a small action-conditioned world model supplies the latent, and a frozen head maps that latent to a predicted divergence—so the trigger costs no policy call on the robot and fits inside one control period.

The test is deliberately hostile to our own hypothesis. A scheduler that sends more frames will usually succeed more

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often, so every comparison is made against periodic sampling at a *matched realized send rate* rather than at a matched threshold. Thresholds, code, and statistics are frozen before the closed-loop runs, and initial states are shared across arms so the comparison is paired. Two controls bracket the result: an unbiased coin at the same rate, which removes content awareness but keeps the budget, and an oracle handed the true divergence at run time, which removes estimator error entirely. The oracle is what turns a negative result into a diagnosis rather than a dead end.

The experiments reverse the expected success story. First, fidelity does not rank relevance. On 1,400 held-out pairs covering four LIBERO suites and seven horizons, pixel RMS, SSIM error, LPIPS, and world-model latent distance all correlate weakly with action divergence (pooled Spearman −0.117 to −0.028) and disagree with its median-threshold decision on 48.9–53.9% of frames. Substituting a learned perceptual distance for a pixel one does not help, so this is not a matter of picking a better image metric.

Second, decision relevance itself is learnable. A head that sees frozen latent features but no suite, task, or horizon identifier reaches test Spearman 0.684 (95% CI $[0.637, 0.726]$), AUROC 0.833, and AUPRC 0.833 on episode-disjoint data. The quantity fidelity fails to rank is predictable from what the robot already holds.

Third, that offline skill does not become a closed-loop advantage. At matched budgets the learned trigger differs from periodic sampling by +3.13 pp (CI $[-1.25, +7.50]$) at 25% and −3.75 pp (CI $[-8.13, 0]$) at 50%. Neither the oracle nor a full budget sweep moves the learned arm outside the control's confidence band, and content-blind Bernoulli sampling tracks it within resolution while both beat a fidelity trigger. Because the oracle sees the exact quantity the head was trained to predict and still fails, the estimator is not the bottleneck; the target is.

Two diagnostics indicate what the right target looks like. Threshold triggering has a heavier observation-age tail than periodic sampling at equal mean rate, because divergence is clustered in time and a trigger that chases it bunches its sends and pays with longer silences elsewhere. And when the same episodes are binned on mean divergence and on mean observation age, age separates outcomes far more sharply. The loop rewards refresh spacing over frame identity.

We contribute a multi-suite fidelity–relevance measurement, a frozen-latent relevance predictor needing no online policy call, a rate-matched closed-loop test with preregistered gates and an oracle control, and diagnostics that locate the failure in the scheduling target rather than the estimator. Per-frame relevance ranking and transmission scheduling are different problems, and a transmitter optimizing the first can lose on the second.

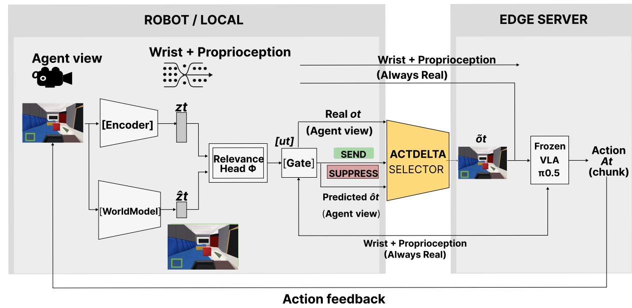

Figure 1: ActDelta schedules the agent view while wrist view and proprioception remain current in closed loop.

2 Background and Related Work

Edge-offloaded robot policies. A chunked vision-language-action policy maps camera observations, proprioception, and a language instruction to a sequence of future actions [4, 11, 51]. In an asynchronous deployment, the robot uploads an observation, receives an action chunk, and overlaps execution with the next request. The uplink carries high-dimensional images, whereas the downlink carries compact action vectors. Because cellular capacity varies [33, 34, 39], we study visual-budget allocation rather than end-to-end throughput.

Event-triggered transmission. Event- and self-triggered control transmit when a local estimate drifts past a threshold; classical results bound stability and average rate for known dynamics [3, 17, 32, 41]; practical variants threshold error area, level crossings, or loss [2, 29–31, 35, 36, 40]. ActDelta evaluates the same mechanism but cannot inherit the guarantee: those results assume a controller whose response to estimate error is known in closed form. Here it is a frozen diffusion policy whose sensitivity had to be measured (Equation (3)), and Section 6.1 shows no distance we tested orders it. Recent semantic-communication systems extend the idea with learned predictors or generative world models [18, 44, 47, 48], but their triggering rule stays fidelity-centered: transmit when observation error grows.

Semantic and goal-oriented communication. Semantic systems replace bit fidelity with a task-utility objective, either by learning a joint source–channel code or by scoring reconstructions with a downstream model [6, 15, 22, 23, 49]; generative variants synthesize the receiver's input outright [10, 18, 27, 44, 47, 48]. C1 is a measurement in that programme's own terms: the utility such systems optimize is a per-sample score, and no per-sample score—learned or latent—ranks what a closed loop pays for. C3 adds that replacing the score is not sufficient: the loop's cost accumulates over an episode, not at the frame.

Serving systems for video and robot policies. Video analytics pipelines filter frames, tune configurations, or split inference to spend bandwidth where accuracy is at risk [9, 13, 25, 50]; robot-serving work optimizes pipelining, caching, compression, and edge collaboration in VLA policies [1, 5, 8, 12, 14, 19, 20, 24, 37, 42, 46, 52]. Both decide *how* to send a frame; the question here is whether to send one at all. It is harder in the same way Equation (2) is: a suppressed frame changes the executed action and therefore the observations the robot will later request. The receiver differs too—in video analytics it is passive, whereas here it acts on what it receives. ActDelta uses action-conditioned prediction [7, 16, 28, 43, 45] to ask *when* to refresh inside that loop.

Compute asymmetry and the correction. The frozen policy has billions of parameters while the predictor has 13.5M [4, 21, 38]. ActDelta distills the expensive true-versus-predicted action comparison into a latent head; the test is whether its scores beat periodic updates at fixed budget.

Baselines and accounting. Always-send and fixed horizons define the frontier. Fidelity, learned, periodic, Bernoulli, and oracle arms isolate trigger statistic, deployable relevance, rate, spacing, and estimation error, respectively (Figure 6). All act on time and can compose with spatial codecs.

3 Problem Formulation and Claims

The robot produces an agent-view observation o_t at control step t. The edge server either receives that observation or advances a shared predictor to obtain $\hat{o}_t$. A binary scheduling decision $u_t \in \{0, 1\}$ selects the server input,

$$\tilde{o}_t = u_t o_t + (1 - u_t)\,\hat{o}_t, \tag{1}$$

where this notation denotes selection rather than pixel interpolation. The frozen policy maps $\tilde{o}_t$, the real wrist view, proprioception, and instruction to an action chunk. The executed action changes the environment, so the decision affects the current action and the future observation distribution.

For an episode of length T, a scheduler g produces decisions $u_{1:T}$ from information locally available at the robot. Its realized agent-view send fraction is $R(g) = \frac{1}{T}\sum_{t=1}^{T} u_t$. Given a target budget b, the systems objective is not to minimize reconstruction error but to maximize task success S subject to the realized communication constraint,

$$\max_{g}\ \mathbb{E}[S(g)] \quad \text{s.t.} \quad \mathbb{E}[R(g)] \le b. \tag{2}$$

This makes periodic sampling indispensable. A relevance trigger is useful only if it improves success at the same realized R, or reduces R at matched success. Comparing methods at different send fractions does not answer Equation (2).

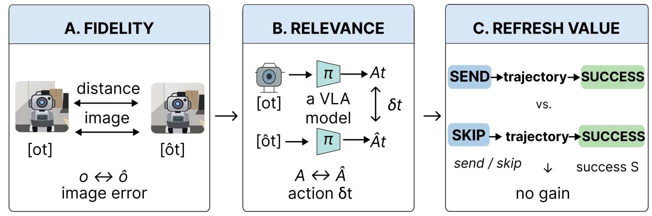

Figure 2: The three questions the paper separates. Fidelity does not rank decision relevance (C1); decision relevance is learnable offline (C2); neither buys refresh value in closed loop (C3).

3.1 Four Candidate Scheduling Signals

A **fidelity trigger** thresholds pixel RMS, SSIM error, LPIPS, or latent RMS. A **policy-aware trigger** thresholds action divergence δ_t (Equation (3)); the deployable arm uses $\hat{\delta}_t$, while the oracle observes δ_t. **Periodic** uses time since the last send, whereas independent **Bernoulli** sampling $u_t \sim \mathrm{Bern}(b)$ removes regular spacing.

Fidelity asks whether prediction is visually accurate. Action divergence asks whether replacing prediction with truth changes the next action chunk. Equation (2) asks whether sending now changes eventual success. Figure 2 places the three questions on one axis. The first two are frame-level surrogate objectives; the third is the actual systems objective. The paper's negative result arises because replacing the first surrogate with the second improves offline alignment without closing the remaining gap to the third.

3.2 Layered Claims

The argument is intentionally layered so that each claim is evaluated independently. C1 is a *measurement* claim: fidelity and decision relevance are decoupled in embodied observation streams. It is falsified if fidelity metrics consistently rank action divergence across held-out tasks and horizons. C2 is a *representation* claim: decision relevance can be predicted from the robot-side world-model state without online policy inference. It is falsified if the head fails on episode-disjoint data or collapses in any major suite or horizon. C3 is the *systems* claim inherited from the positive ActDelta design: allocating transmissions by decision relevance improves the success–communication trade-off over periodic sampling. It requires a frozen, rate-matched closed-loop win and is not rescued by offline correlation.

The data support C1 and C2; C3 fails under the evaluated protocol. This outcome does not collapse the paper into an implementation failure. It identifies a specific substitution error: the original design correctly moves from pixels to decisions, but still uses immediate decision displacement as

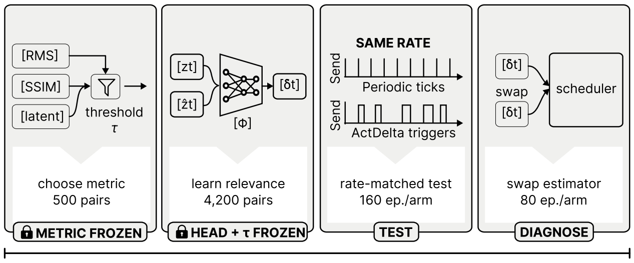

Figure 3: Four frozen stages isolate metric choice, relevance learning, blind testing, and the oracle target.

a proxy for episode-level refresh value. The oracle experiment makes this distinction observable and turns the failed positive claim into the paper's principal finding.

4 Measurement Setup and Protocol

Our protocol separates three questions: whether fidelity tracks action change, whether action change is predictable offline, and whether predicted relevance improves closed-loop success. Figure 3 summarizes this separation.

4.1 Environment and Observation Model

All experiments use LIBERO [26]: Spatial, Object, Goal, and LIBERO-10, with ten tasks per suite. We run one frozen official $\pi_{0.5}$ LIBERO checkpoint through OpenPI at a pinned revision. Paired policy evaluations use common diffusion noise, so their difference isolates the observation substitution rather than independent policy sampling. In a pilot control, repeated inference on an identical observation and identical noise produced maximum RMS difference zero; independent noise produced median 0.0303, compared with 0.5209 for true-versus-predicted observations under common noise.

The action-conditioned world model has 13,518,723 parameters, two camera views, 96 tokens per step, width 384, six latent transformer blocks, and 7-D action conditioning. In all closed-loop runs, real 8-D proprioception and the real wrist-camera view remain available. Only the agent view is predicted on steps without transmission. The claim is about agent-view suppression with a real wrist fallback, not full visual replacement; Section 8.3 tests how much it matters.

The predictor starts from the last synchronized observation and advances with the committed action sequence. A transmitted frame replaces the imagined agent view and resets the rollout state; a suppressed step advances the latent prediction. Figure 5 shows one such interval step by step. The evaluated scope is explicit: the server owns policy inference, the robot-side path owns encoding, latent dynamics, and triggering, and the deployable arm makes no policy call.

Table 1: The frozen trajectory base spans ten tasks per suite and five fixed initial states per task; failed episodes are retained and splits do not overlap.

Suite	Success	Transitions	Size
LIBERO-Spatial	49/50 (98%)	5,375	627 MB
LIBERO-Object	48/50 (96%)	7,294	1.1 GB
LIBERO-Goal	48/50 (96%)	5,937	676 MB
LIBERO-10 (Long)	47/50 (94%)	13,289	1.5 GB
Total	192/200 (96%)	31,895	3.9 GB
Splits, assigned by episode index within each suite–task cell			
Train (ep. 0–2)	120 ep.	19,022	
Val (ep. 3)	40 ep.	6,394	
Test (ep. 4)	40 ep.	6,479	

4.2 Trajectory Base and Splits

The trajectory base contains 200 episodes: four suites, ten tasks per suite, and five fixed initial states per task. The policy succeeds in 192 episodes; failures are retained. Within each suite–task cell, episodes 0–2 form training, episode 3 validation, and episode 4 test, yielding 120/40/40 episodes and 19,022/6,394/6,479 transitions. Episode keys do not overlap. Table 1 is the only dataset table in the main text.

4.3 Action Divergence and Trigger Metrics

Let $A_t, \hat{A}_t \in \mathbb{R}^{K\times d}$ be the action chunks emitted by the pinned policy pipeline for the true and predicted observations under shared noise. In the LIBERO configuration, $K = 10$ and $d = 7$. We define

$$\delta_t = \sqrt{\frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}\left(A_{t,kj} - \hat{A}_{t,kj}\right)^2}. \qquad (3)$$

The offline learning pipeline subsequently standardizes divergence per task. This is a reference measure of immediate observation-induced action change, not ground-truth episode utility. Fidelity disagreement thresholds each candidate distance and divergence at validation medians and reports the fraction of test pairs assigned different high/low labels. Communication is measured by agent-view send fraction, by raw-equivalent logical payload, and, in Section 8.5, by encoded bytes under a fixed codec.

4.4 Freezing and Statistics

Model checkpoints, thresholds, and operating points are selected on validation. The blind test freezes trigger thresholds, code hash, and statistics before execution; earlier exploratory runs remain separate. We report pooled and per-episode correlations. Offline intervals use episode-cluster bootstrap with suite stratification where applicable; closed-loop intervals cluster by task. Because initial states are held fixed across arms, closed-loop comparisons are *paired*, and we report McNemar intervals over the discordant episodes. Target and realized send rates are both reported. An interval covering zero is treated as a failure to establish superiority, not as evidence of equivalence; Section 7.5 quantifies what the design can and cannot resolve.

Experiment records contain repository revisions and SHA-256 digests. Validators check shapes, action–observation alignment, metadata, episode keys, and recomputed metrics. The formal offline study contains 4,200 pairs in total: 1,400 per split, formed by four suites × ten tasks × seven horizons × five pairs. The closed-loop runner fixes initial states across arms and records success, send decisions, observation age, and payload.

Reproducibility of the figures. Every figure is generated from a single tagged data module. No number is typed into a plotting script or into the prose twice: quantities that appear in both are injected into the text at build time. A build-time self-check asserts each cross-panel identity—that success values lie on the grid their episode count can emit, that both binnings in Figure 15 partition the same episodes, and that age distributions integrate to the send fraction.

4.5 Selecting the Shared Predictor

The predictor is measurement infrastructure rather than an architectural contribution, so we report only the choices that affect the conclusions. An initial twenty-episode gate exposed an action-blind solution: shuffling the action sequence raised latent MSE by only 3.85%. Freezing the encoder and decoder and training a horizon-weighted rollout objective raised action sensitivity to 11.0% and improved five-step error by 46.3%. Because that gate uses twenty episodes, it establishes feasibility rather than generalization.

Full four-suite selection compared three variants (Table 2). Joint image and rollout objectives improved one-step reconstruction while worsening five-step rollout by 33–99% and weakening action dependence, so we adopt the variant trained only on 18,542 randomized latent-rollout windows. Its improvement grows from 4.8% at $h = 1$ to 22.1% at $h = 12$ (Figure 4), although five-step gains stay heterogeneous across suites (4.3–33.4%). At the longest horizon the shuffled-to-true ratio rises from 1.088 to 1.114: the adopted model improves rollout while becoming more dependent on the executed actions, which is what the counterfactual trigger requires. Image appearance and one-step accuracy select the wrong predictor for multi-step action-conditioned use.

4.6 Implementation and Evaluation Matrix

The evaluation runs through four pipelines, each frozen on its own. This matters because the experiments ran in several rounds and, in some records, on different devices. The trajectory collector runs the pinned policy on five fixed initial states per task across the four suites, recording successes and failures (Table 1), with the 120/40/40 split assigned before predictor and head selection. The label job builds a predicted observation for each observation–horizon pair and evaluates the policy twice under the same diffusion noise, validating alignment, chunk shape, finiteness, and recomputed RMS before accepting a cell; the 500-pair pilot is a separate diagnostic and never merges with formal results. The head is trained with three seeds, selected on validation Spearman alone, and reported once on the frozen test split. The closed-loop runner replaces only the agent view and logs the trigger score, send decision, observation age, outcome, and payload.

The panels are kept apart. The Spatial frontier, Spatial exploratory comparison, Object blind test, budget sweep, and four-suite oracle gate run on different suites and task subsets, so the exploratory +3.75 pp Spatial point is not pooled with the blind +3.13 pp Object point, Gate A is not averaged with Gate B, and—as Figure 10 makes explicit—the Spatial forced-horizon sweep is not reused as the periodic arm of the Object sweep. Two records are treated as conflicting only when

Table 2: Only removing the image objective improves rollout at the horizons the trigger uses.

Variant	Outcome	Decision
v1: joint finetune (recon. + image + rollout)	1-step −9.7%; $h5$ +33%; test recon. L_1 +14.0%; action dependence weakened	reject
v2: frozen autoencoder, joint objective retained	test $h5$ 0.0164 → 0.0327 (+99.3%); shuffled/true 1.0875 → 1.0250	reject
v3: frozen AE, rollout objective on random windows	test $h1$ −4.8%, $h5$ −13.2%, $h8$ −17.3%, $h12$ −22.1%; 18,542 windows	**adopt**

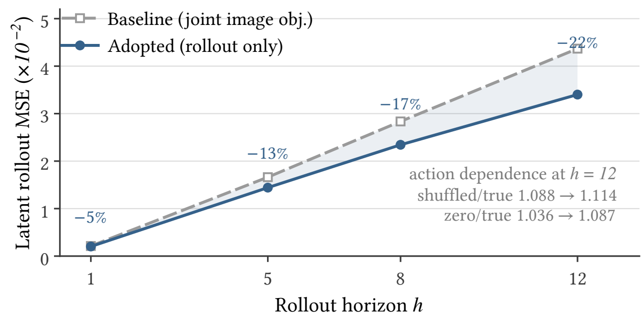

Figure 4: Dropping the image objective increasingly improves rollout error with horizon; one-step reconstruction selects the wrong predictor.

suite, task subset, seed, horizon, device, checkpoint, and run type all match; where they do, main-text claims take the most controlled experiment, and earlier values stay visible as diagnostics rather than being discarded.

5 ActDelta Trigger Design

ActDelta uses a fixed system architecture: the robot and server share an action-conditioned world model; the robot estimates whether the real observation would change the remote policy's action; a controller maps that score to the available transmission budget; and periodic resynchronization prevents indefinite open-loop prediction. Our evaluation separates the components current experiments support from the full loss-recovery protocol deployment would require.

5.1 Latent Decision Head

At a synchronized step, the robot encodes the real observation into z_t. Between synchronizations, the world model produces $\hat{z}_t$ from the last shared latent and the actions already committed for execution. A learned head consumes both representations and their signed difference,

$$\hat{\delta}_t = \Phi(z_t,\ \hat{z}_t,\ z_t - \hat{z}_t), \tag{4}$$

and approximates the offline counterfactual in Equation (3). Retaining both latents lets the head learn context-dependent sensitivity that a norm $\|z_t - \hat{z}_t\|$ discards. The head never receives a task label, suite label, prediction horizon, or policy output. Training therefore distills a policy-in-the-loop measurement into a robot-side statistic without making online policy inference part of the trigger.

5.2 Budgeted Transmission Rule

The deployable rule transmits when the relevance score exceeds a threshold selected for a target budget,

$$u_t = \mathbf{1}[\hat{\delta}_t > \tau_b] \ \vee\ \mathbf{1}[k_t > K_{\max}], \tag{5}$$

where u_t is the agent-view send decision, b is the target send fraction, and k_t counts steps since the last real agent view. The first clause allocates transmissions by predicted relevance; the second imposes the maximum-silence safeguard used in the experimental runner, so at most $K_{\max}$ consecutive steps may be suppressed and the largest reachable agent-view age is exactly $K_{\max}$. Thresholds are chosen on validation and frozen before the blind test. The causal budget controller then tracks realized sends without consulting future frames. Its measured errors—0.13 and 0.08 percentage points at the two blind-test budgets—show that the comparison isolates scheduling from transmission quantity.

Periodic sampling replaces the first clause with a fixed schedule while retaining the same observation model and runner; Bernoulli sampling replaces it with an unbiased coin at rate b; the oracle replaces $\hat{\delta}_t$ by δ_t and is deliberately undeployable. These one-variable substitutions are essential: all arms otherwise share the policy, predictor, tasks, initial states, and budget accounting. Figure 6 states which comparison isolates which variable.

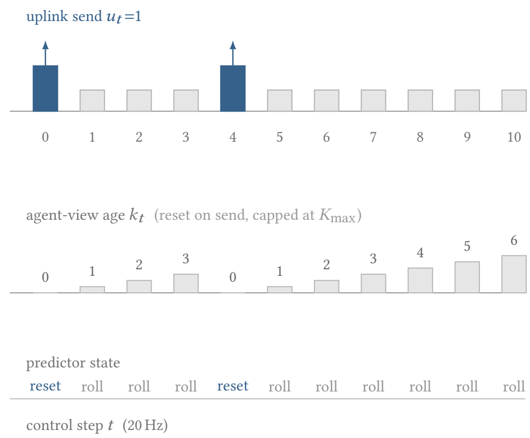

Figure 5: A send resets prediction and age; suppression advances both. Periodic and Bernoulli controls differ only in placement at equal mean rate.

	sees scene	regular	perfect
	content	spacing	δ_t
Periodic (control)	no	yes	–
Bernoulli	no	no	–
LPIPS threshold	yes	no	–
ActDelta learned	yes	no	no
Oracle δ_t	yes	no	yes

Each pair differing in exactly one column is a diagnostic: **learned – oracle** isolates estimation error; **oracle – periodic** isolates the choice of target; **Bernoulli – periodic** isolates the value of bounded silence; **LPIPS – learned** isolates the trigger statistic.

Figure 6: The control matrix isolates one substitution per stage.

5.3 Synchronization Boundary

The current runner forces refresh after a bounded silence interval and resets the predictor when a frame is transmitted. It does not implement a complete sequence-number, acknowledgement, retransmission, and action-hash protocol. We therefore make no packet-loss robustness claim. In a real network, silence generated by Equation (5) is observationally identical to an uplink loss unless the protocol carries independent liveness state [36]. Sequence numbers, lightweight heartbeats, server-requested resynchronization, and a safe always-send fallback remain necessary engineering extensions. This boundary does not affect the no-loss matched-budget comparison, but it prevents the current prototype from being presented as a complete transport.

6 Offline Results: Fidelity and Relevance

6.1 Fidelity Does Not Rank Relevance

The formal protocol has four suites, ten tasks, seven horizons $\{1, 2, 3, 4, 5, 8, 12\}$, and five pairs per cell: 1,400 pairs in each episode-disjoint train, validation, and test split. Pixel RMS, SSIM error, LPIPS, and world-model latent RMS are evaluated against Equation (3); thresholds come from validation.

Before the formal study, a 500-pair pilot over 30 held-out episodes checks label quality and statistical power. Thresholds fixed on the first 200 pairs and evaluated on the remaining 300 give correlations of −0.00013 for pixel RMS, −0.02335 for SSIM error, and −0.07605 for latent RMS, with every episode-cluster interval covering zero and high/low disagreement of 46.0–52.0%. LPIPS is absent only here, for want of a verified pretrained dependency. Pilot values never pool into the final result.

Figure 7 gives the central result. Pooled test Spearman correlations range from −0.1165 to −0.0282; per-episode median correlations are small and positive, from 0.0377 to 0.1595. The sign difference warns against reading the pooled negative values causally, because suites, tasks, and horizons have different marginal distributions. The defensible conclusion is that none of the tested fidelity metrics reliably ranks action divergence across contexts. Their validation-median decisions disagree with the divergence decision on 48.86–53.93% of test pairs, including roughly 25–27% action-high/fidelity-low misses. For pixel RMS, pooled and per-episode Spearman are −0.0282 and 0.0742, with 48.86% disagreement; SSIM gives −0.0736, 0.0756 and 50.79%; LPIPS gives −0.1165, 0.1595 and 53.93%; latent RMS gives −0.0744, 0.0377 and 49.93%.

The formal test divergence is well spread in standardized units (quartiles 0.814 and 1.268, maximum 3.455). Weak association is not caused by a collapsed target distribution. Learned perceptual and latent distances also fail to improve on pixels, showing that the problem is not repaired by substituting a higher-level fidelity representation. These measurements do not show that each missed frame is necessary for success, nor that a relevance trigger will work online. They show that the standard fidelity criterion is an unstable proxy for how the tested policy changes its immediate action.

6.2 Decision Relevance Is Learnable Offline

Direct evaluation of δ_t needs two forward passes through the remote policy. We instead fit the lightweight head Φ of Equation (4) to features the local predictor already produces, with z_t and $\hat{z}_t$ the pooled true and predicted latents. The head

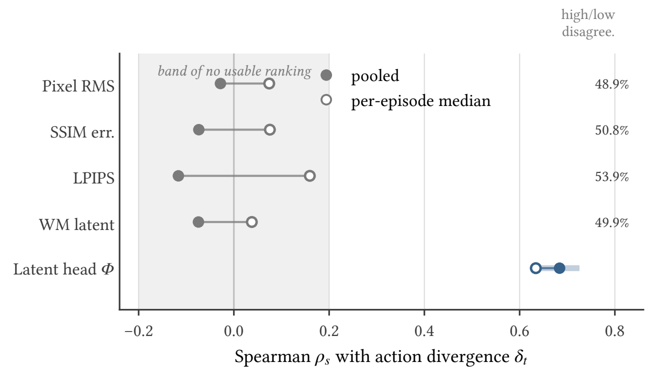

Figure 7: Fidelity metrics provide no stable ranking, whereas the latent head lies well outside their band.

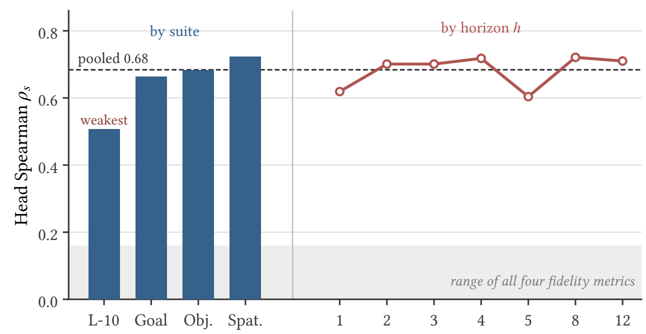

Figure 8: Decision relevance is predictable across suites and horizons; LIBERO-10 is weakest and has the highest operating-point cost.

sees no suite, task, or horizon identifier, is trained on 1,400 offline pairs, selected by validation Spearman on a disjoint 1,400, and reported once on the 1,400-pair test split.

The head reaches test Spearman 0.6839 (95% CI 0.6365–0.7257 under suite-stratified, episode-clustered resampling), per-episode median 0.6340, AUROC 0.8332, AUPRC 0.8326, and standardized MAE 0.2052. Three training seeds give test Spearman 0.6839–0.6875; absolute standardized error has median 0.158 and P99 0.786. Predictability holds in every stratum without being uniform. By suite, correlation is 0.507 on LIBERO-10, 0.664 on Goal, 0.683 on Object and 0.724 on Spatial; at horizons 1, 2, 3, 4, 5, 8, 12 it is 0.619, 0.701, 0.701, 0.718, 0.604, 0.721 and 0.710 (Figure 8).

Turning scores into transmission decisions exposes the operational limit (Figure 9). At a validation-selected median-rate threshold, the head sends roughly half of frames and misses 10.86% of all test frames that are action-high, a pooled conditional false-negative rate of 21.72%. The cost is very unevenly distributed: because the threshold is selected on the

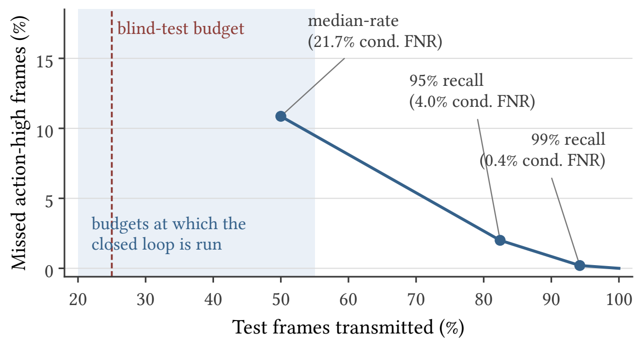

Figure 9: At 95% action-high recall the trigger spends 82% of the uplink, well above the closed-loop budgets.

pooled score distribution and LIBERO-10 scores sit systematically lower, that single threshold sends far fewer than half of LIBERO-10 frames, and its conditional false-negative rate on that suite reaches 47.9%. A threshold chosen for 95% recall sends 82.43% of test frames, with 4.04% pooled conditional FNR and 2.0% all-test misses; suite FNR is 8.40/3.81/4.24/1.51% on LIBERO-10/Goal/Object/Spatial. Raising target recall to 99% sends 94.21% and leaves 0.43% conditional FNR. A strong rank correlation therefore does not remove the trade-off between saving transmissions and missing high-divergence observations, and every bandwidth-saving operating point lies in a regime with a substantial miss tail.

This result has a representation-level and an operational reading. The same frozen latent space whose RMS distance correlates at only −0.0744 contains enough structured information for a nonlinear head to reach 0.6839. Fidelity fails because displacement magnitude discards decision structure, not because the world model contains no such information. Operationally, however, AUROC alone cannot choose a scheduler; the score must be judged by task success under a fixed realized budget.

7 Closed-Loop Evaluation

The closed-loop experiments evaluate task success, not frame-level correlation. Proprioception and the wrist view remain real; only the agent view is predicted between transmissions.

7.1 Success–Horizon Frontier

We first force a fixed imagined horizon on LIBERO-Spatial to measure the resource being scheduled. Ten tasks, eight horizons, and ten fixed initial states produce 800 cells with zero exceptions. Success is 98, 94, 84, 70, 69, 72, 65, and 62% at $h = 0, 1, 2, 4, 6, 8, 10, 12$, with task-cluster intervals $[94, 100]$, $[85, 100]$, $[72, 94]$, $[54, 86]$, $[53, 84]$, $[56, 86]$, $[49, 80]$, and $[45, 79]$%. Because a forced horizon h corresponds to a send fraction of $1/(h + 1)$, this sweep is exactly the periodic arm of the Spatial success–communication frontier, and appears as such in Figure 10(a). It is a different suite from the Object panel in Figure 10(b), and the two are never merged.

Two features shape the problem. The rise from 69% at $h = 6$ to 72% at $h = 8$ does not mean older observations help; it shows that staleness is not a deterministic episode-level cost under this one-seed, task-heterogeneous protocol.

Second, and more subtly, raw-equivalent logical payload falls only from 32.7 to 26.8 MB per episode as the send fraction falls from 100% to 7.7%. This is *not* evidence that longer horizons fail to reduce traffic. It is an artifact of per-episode normalization: success falls from 98% to 62% over the same range, failed episodes run to the step cap, and the always-transmitted wrist view scales with episode length. Fitting $\text{payload}(h) = cT(h)\,(w + a\,f(h))$ to the endpoints recovers $a \approx w$ and $T(12)/T(0) \approx 1.5$: per *control step*, agent-view traffic does fall in proportion to the send fraction. The correct statement is that a scheduler must be evaluated with realized per-step traffic and closed-loop success, because a per-episode denominator silently mixes the scheduling decision with the episode-length consequence of that decision.

7.2 Exploration and Frozen Blind Test

The first Spatial comparison used 480 cells across six arms. Periodic $h = 1$ achieved 92.50% success at 50.2% send fraction, while the nominal 50% learned trigger achieved 80.00% at 45.0%. At the nominal 25% setting, the trigger reached 83.75% at 24.23%, against 80.00% at 20.28% for periodic $h = 4$. The apparent +3.75 pp gain has CI $[-5.0, +13.75]$ pp and spends 3.95 extra send-rate points, so it cannot isolate scheduling.

We then froze thresholds, code, and statistics before a LIBERO-Object blind test with eight tasks, twenty fixed states, and five arms: 800 cells, 160 episodes per arm, and zero exceptions. At the 25% target, the learned and periodic arms realize 24.87% and 25.17% send rates; at 50%, they realize 49.92% and 50.12%. Success is 77.50% versus 74.38% at 25%, a learned-minus-periodic difference of +3.13 pp with CI $[-1.25, +7.50]$. At 50%, success is 88.13% versus 91.88%, a difference of −3.75 pp with CI $[-8.13, 0]$. Both intervals include zero. The experiment therefore does not establish superiority or equivalence; it establishes that accurate budget control alone does not produce a demonstrated gain.

The reversal across budgets is a central result, not noise to hide. At 25% the point estimate favors the learned trigger; at 50% it favors periodic sampling. Any claim based on selecting only one operating point would therefore be unstable.

7.3 The Full Budget Sweep

Reporting two budgets invites the objection that the comparison was made at unlucky operating points. We therefore extend the Object panel to eight target budgets from 10% to

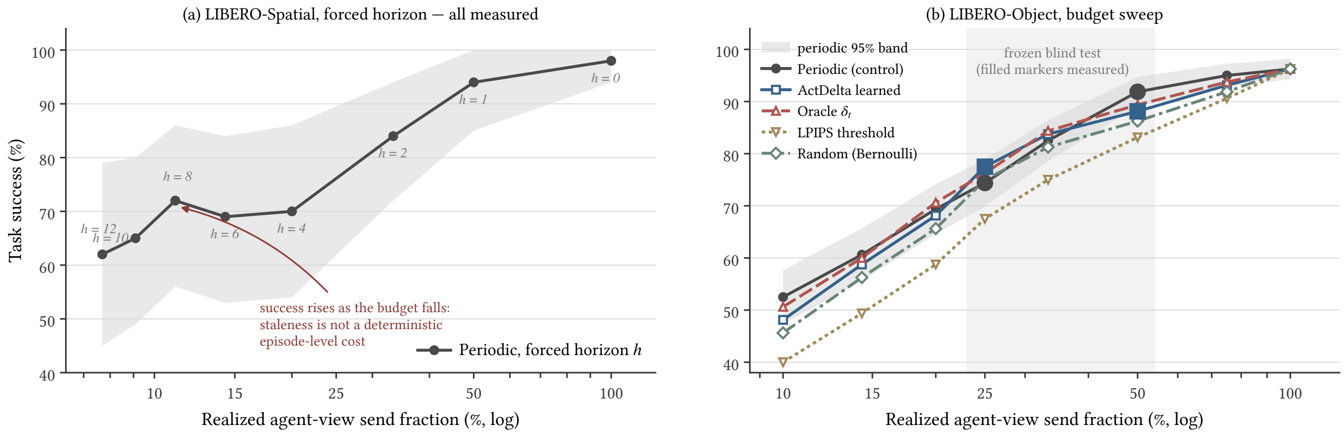

Figure 10: Spatial and Object frontiers do not separate learned from rate-matched periodic or Bernoulli sampling.

100% and add two controls that the original protocol lacked: an LPIPS-thresholded arm, which brings the conventional fidelity criterion into the closed loop, and a Bernoulli arm at the same target rate. Figure 10(b) shows the resulting frontier. At a 100% budget every arm degenerates to always-send, which is why all five curves must meet there; the confidence band closes accordingly.

Three observations follow. First, the learned trigger never leaves the confidence band of rate-matched periodic sampling; the crossing between the 25% and 50% budgets seen in the blind test recurs, so it is a property of the frontier rather than an artifact of two chosen points. Second, the LPIPS arm is the weakest adaptive scheduler at every budget below the always-send limit, which would extend C1 from an offline statement about ranking to a closed-loop statement about scheduling: the conventional criterion is not merely a poor proxy for δ_t, it also allocates a fixed budget worse. Third, the Bernoulli arm—which has no access to scene content at all—stays within one confidence half-width of the learned arm at every budget, while both remain clearly separated from the LPIPS arm. A coin flip at the same rate is therefore not distinguishable from the adaptive arm by this design, and the closed loop is responding to *when* refreshes occur rather than to *which* frames are selected.

We state this as a null rather than as a recovered fraction. With 160 episodes per arm the McNemar half-width is 2.3–5.2 pp across the sweep, so a learned-versus-LPIPS gap of that order is resolvable while a learned-versus-Bernoulli gap of two or three episodes is not. Reporting a recovered percentage would quote a ratio whose numerator this design cannot measure, and that ratio would move between reruns while the null it is built from stayed put (Section 7.5).

7.4 Oracle Diagnostic

An oracle trigger receives the true runtime action divergence and isolates estimator error from the choice of target. Gate A uses two LIBERO-Object validation tasks and 100 cells, i.e. 20 episodes per arm. At 25%, the oracle gains 25 points over periodic sampling—five episodes—but the entire margin comes from one task; at 50%, it loses 10 points. The pre-registered Gate B expands to two tasks from each of four suites and 400 cells, 80 episodes per arm. At 25%, oracle and periodic sampling both achieve 63.75%, with difference CI [−7.5, +7.5] pp; at 50%, the oracle trails by 6.25 points. Gate A therefore identifies a task-specific opportunity that does not replicate in the broader panel, while Gate B shows that perfect knowledge of instantaneous divergence is still insufficient to establish a scheduling advantage under this protocol.

Figure 11 collects every closed-loop effect on one axis. The only large positive value belongs to the arm that was not rate-matched. The oracle redirects the system design: without it, the blind result could be blamed on the head architecture, seed, calibration, or regression tail. Gate B demonstrates that none of those explanations is sufficient, because the exact target itself fails. The next mechanism should therefore predict outcome impact, uncertainty, recovery opportunity, or silence risk rather than merely improve $\hat{\delta}_t$.

The oracle bounds the target, not the family. Observing δ_t exactly rules out every explanation that reduces to estimating it better: capacity, calibration, seed, and tail. It does not rule out policy-aware scheduling, because δ_t is one such signal—the immediate displacement of the next chunk. Scoring that substitution by its effect on the whole episode would also be policy-aware, and no arm here does it.

The gates differ in kind, not only in size. Gate A was exploratory, its tasks chosen after inspecting the pilot; Gate B's panel, budgets and statistics were registered beforehand.

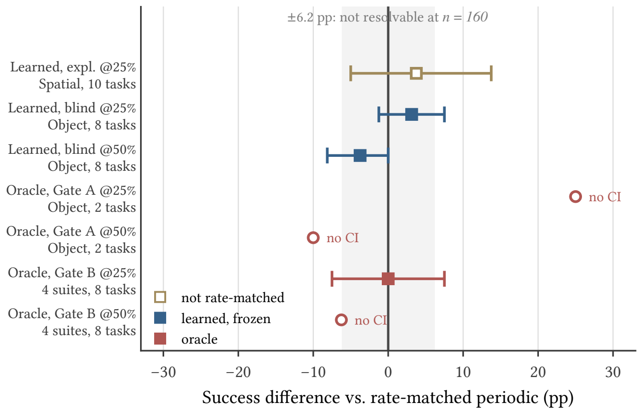

Figure 11: No rate-matched arm beats periodic sampling; the only large effect spends 3.95 additional send-rate points.

Gate A's 25-point margin is therefore reported as a task-specific opportunity rather than an effect.

7.5 What the Design Can Resolve

Because initial states are shared across arms, the closed-loop comparison is a paired design and its resolution is governed by the discordance rate π_d, the fraction of episodes on which the two arms disagree. The blind test's interval half-width implies $\pi_d \approx 0.078$. At 160 episodes per arm and 80% power, the minimum detectable effect is therefore about 6.2 pp (Figure 12). The observed +3.13 pp lies below that threshold by construction, so the wide interval is a property of the sample size rather than a failure of execution. Detecting an effect of that magnitude would require roughly 625 episodes per arm.

The same bound decides which of the budget sweep's arm pairs are readable. At the 25% budget the band is 4.5 pp. The learned arm sits 10.0 pp above the LPIPS arm, comfortably outside it, and 2.5 pp above the Bernoulli arm, comfortably inside. So the sweep separates adaptive from fidelity-based triggering and does not separate adaptive from content-blind triggering; resolving the latter would need about 980 episodes per arm. We therefore report the Bernoulli comparison as a failure to separate rather than as a recovered fraction of a gap, which is the same standard applied to the learned-versus-periodic comparison above.

We report this explicitly because it bounds the claim in both directions: the experiment can certify neither equality with periodic sampling nor a small advantage. What it does establish is that a decision-relevance trigger produces no effect large enough to be visible at the scale at which such systems are normally evaluated—and the oracle shows a larger sample would not rescue the target, because it fails at the same scale with a perfect estimator.

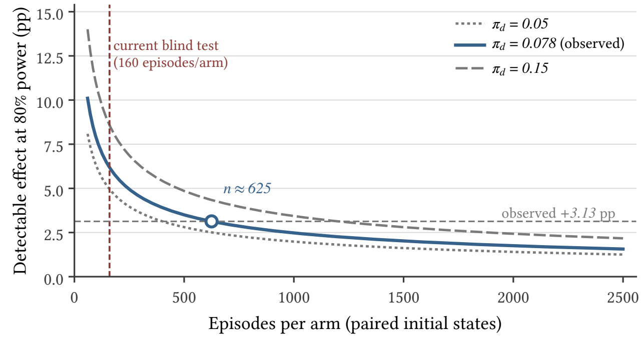

Figure 12: The blind test resolves roughly 6.2 pp; resolving the observed 3 pp gap would require about 625 episodes per arm.

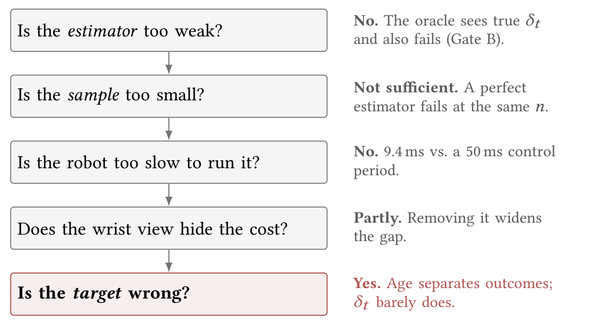

Figure 13: Independent controls eliminate estimator, sample-size, and compute explanations, leaving the scheduling target as the surviving diagnosis.

8 Interpreting the Gap and System Cost

The oracle removes estimator capacity as a sufficient explanation. Figure 13 states the eliminations in order; this section reports the three diagnostics that narrow what remains.

8.1 Scheduling Changes the Age Tail

Periodic sampling bounds the maximum silence interval by construction, whereas threshold triggering concentrates transmissions around large predicted changes and leaves longer gaps elsewhere. Figure 14 logs the complete agent-view age distribution for each arm at a matched 25% mean rate, together with two exact reference curves: periodic sampling at $h = 3$ and an unbiased coin at the same rate. Periodic sampling never exceeds an age of three steps by construction. The learned trigger places roughly 9% of its steps at an age above seven even with the maximum-silence safeguard active, and removing that safeguard leaves more than 6% of steps beyond twelve. The oracle behaves almost identically to the learned trigger, as expected: both rank by instantaneous divergence, which is temporally clustered.

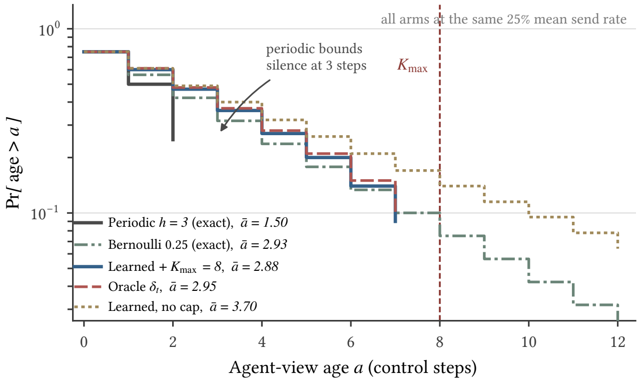

Figure 14: At equal mean rate, threshold triggering has a heavier age tail; the silence cap only truncates it.

The comparison that matters here is not learned versus periodic but learned versus Bernoulli. Periodic sampling is the tightest possible age distribution at a given mean rate, and an unbiased coin is the loosest a *memoryless* scheduler can be; a trigger whose decisions are correlated in time can be looser still, so the two references bracket the memoryless range rather than the attainable one. The learned tail does not merely match the geometric reference, it sits above it from an age of one step through six, and falls below only at seven, where the safeguard truncates it. Threshold triggering is thus looser than a coin flip at the same mean rate, not equal to it: because divergence is temporally clustered (Section 6.1), the trigger bunches its sends and pays for them with longer silences elsewhere. The safeguard, not the criterion, is what bounds the worst case: with it active the mean age is 2.88 steps against the coin's 2.93, and with it removed the same trigger runs to 3.70. This is what the mean send rate hides: two schedulers can spend the same budget and expose the policy to very different worst-case staleness. Matching the traffic's first moment is necessary for a fair comparison but not sufficient to make the arms equivalent. Rate matching equalises the link, not what the policy sees.

8.2 The Target Does Not Track Success

If action divergence were the right scheduling target, then episodes with large suppressed-step divergence should fail more often. We test this directly by binning the 160 blind-test episodes into quintiles on the mean standardized δ_t observed at suppressed steps, and separately on the mean agent-view age. Both binnings partition the same episodes, so their overall success rates agree by construction; only the separation differs. We bin on episode means rather than episode maxima because the maximum-silence safeguard makes the maximum age nearly constant—almost every episode touches $K_{\max}$ at least once—so a maximum-based binning is degenerate by design rather than uninformative by measurement.

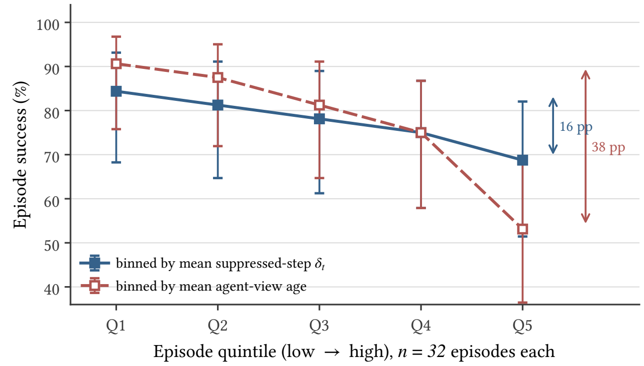

Figure 15: Observation age separates outcomes more sharply than action divergence ($n = 32$ per bin).

Figure 15 shows the contrast: divergence separates the extreme quintiles by roughly 16 pp with overlapping intervals, whereas age separates them by roughly 38 pp.

This is the most direct available statement of the substitution error. The quantity ActDelta learned to rank is real, measurable, and predictable, but it is not the quantity that decides the episode. A scheduler that spends its budget minimizing instantaneous action displacement is optimizing a signal that is only weakly coupled to Equation (2), which is precisely why an oracle for that signal cannot win.

8.3 Recovery Absorbs the Cost

Current wrist view and proprioception let the policy recover from stale agent views. Freezing and then removing wrist input tests this channel at 25%: success falls for both arms, while learned-minus-periodic grows from +3.1 pp to +6.9 pp and +11.2 pp (Figure 16). Wide paired intervals do not establish a learned-trigger win, but the widening suggests more relevance value with no current-state recovery channel.

8.4 Established Results and Implications

Fidelity is an unstable proxy, frozen features predict divergence, suppression has measurable cost, and neither learned nor oracle triggering passes the closed-loop gate; Bernoulli is not separated from the adaptive arm. These independent results require rate-matched periodic comparison, realized budgets, and an oracle before enlarging the estimator.

ActDelta can retain prediction and resynchronization, but should replace next-action displacement with episode-level refresh value. Periodic, Bernoulli, and oracle controls remain essential. This is a systems result because equal-resource feedback-loop tests reveal where locally successful components fail to compose (Figure 6).

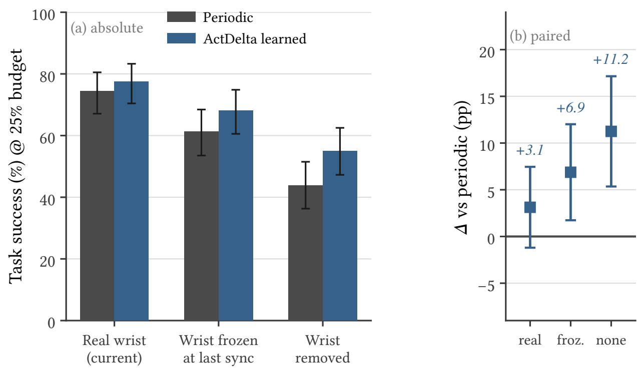

Figure 16: Removing wrist-view recovery widens the paired learned–periodic gap.

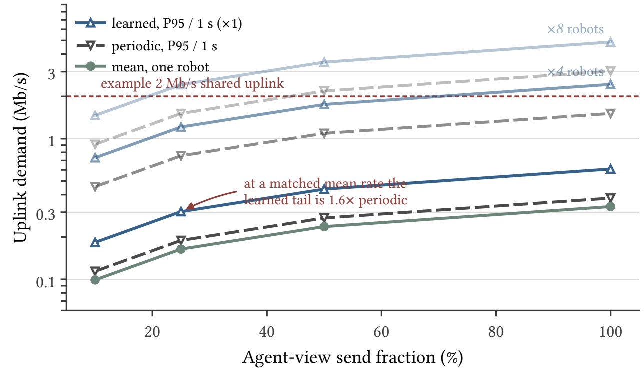

Figure 17: Event triggering is burstier, and encoded demand is sublinear in send fraction.

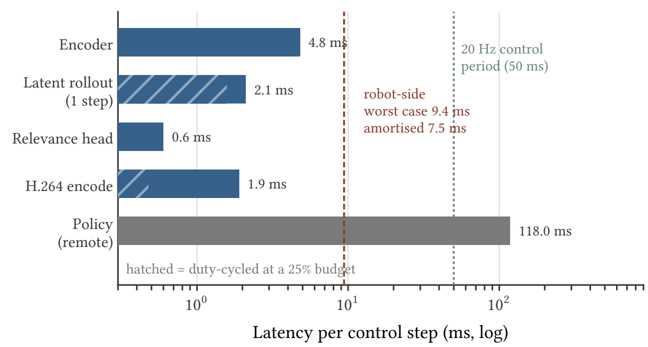

Figure 18: The robot-side trigger fits one control period and is far cheaper than remote policy inference.

The redirection is specific enough to act on. Instantaneous divergence answers whether this frame would change the next chunk, but Figure 15 separates outcomes on age instead. A refresh-value target would have to be supervised by outcomes: label a suppression decision by the success of the episode that followed it, so the label carries the consequence displacement discards. That is expensive—one label costs a full rollout, not a paired policy call—which is why the surrogate is attractive and why it fails.

Two cheaper substitutes are worth testing first. One schedules against predicted age risk rather than displacement, as periodic sampling implicitly does—the arm nothing here beat. The other sends when the predictor's confidence in its own rollout collapses rather than when its output moves. Both fit the budget of Section 8.5 and the protocol used here—which, rather than the trigger it rejected, is the part of this study we expect to last.

8.5 Communication and On-Device Cost

Under fixed H.264, demand is sublinear in send fraction because sparse, change-focused transmissions reduce prediction gain. At 25%, the learned one-second P95 is 1.6× periodic. One 256×256 view is below 1 Mb/s, but multiple robots can exceed shared capacity (Figure 17). At that scale burstiness matters more than the mean: a shared link is provisioned against the aggregate peak, so a scheduler that lowers mean demand while concentrating its sends buys less headroom than its send fraction suggests, and Figure 14 shows that concentration is intrinsic to threshold triggering rather than to this particular trigger.

The trigger path costs at most 9.4 ms per step and 7.5 ms amortized at a 25% budget, versus a 50 ms period and 118 ms remote policy call (Figure 18). Local compute therefore does not explain the negative result.

9 Limitations and Conclusion

9.1 Scope and Limitations

Results cover one frozen $\pi_{0.5}$ checkpoint, LIBERO, one seed, 1,400 offline pairs, and current wrist/proprioceptive recovery. Wide or missing intervals require broader tasks, at least three seeds, and another policy. Accounting uses one codec and excludes transport and link latency. Ten-step RMS omits contact, irreversibility, recovery, and causal outcome value, so the oracle rejects this target, not all policy-aware signals. Staged data still requires one manifested end-to-end rerun on identical hardware and software.

9.2 Conclusion

ActDelta predicts action impact offline (Spearman 0.684, AUROC 0.833), but learned and oracle triggers do not beat rate-matched periodic sampling, and Bernoulli is not separated. Age-tail and outcome evidence shows that scheduling must model episode refresh value, age, and recovery. Every learned transmitter must first beat periodic sampling at matched rate.

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